MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)

UG Programme (Honors) Degree October 2025 Regular Examinations

First Semester

251CCRUBC101 Digital Fundamentals

Time: **2 Hours** Maximum Marks: **40**

Answer one question from each bunch

BUNCH I

I.A A digital weighing scale represents weight in BCD for ease of display. The current reading is 59 , and an additional weight of 28 is added. Calculate the total weight using BCD addition and describe how the scale corrects invalid results. (CO1:8 Marks)

OR

I.B A digital display system shows numerical data using Binary Coded Decimal (BCD). The current display shows 45 (decimal).

i. Represent the number in BCD format. (2 marks)
ii. Add 27 to the display value using the BCD addition method, showing all steps. (6 marks) (CO1:8 Marks)

BUNCH II

II.A

II.A.1 Apply **De Morgan's Law** to the complement of the expressions:

i) ((A+B+C)D)'
ii) ((A+BC')'+D(E+F')')'

(CO2:8 Marks)

OR

II.B Rahul, a smart home automation engineer, is designing a control system to manage various devices like lights, fans, and alarms. To implement the logic for device operations, he needs to use basic logic gates such as AND, OR, and NOT. For testing and documentation purposes, he must create truth tables for each gate and draw their graphical representations.

II.B.1 Your task is to help Rahul by providing the truth tables and graphical representations for the logic gates AND, OR, NAND and NOR to assist in the smart home control system design.

(CO2:8 Marks)

BUNCH III

III.A The Boolean function **F(X, Y, Z) = XY + X'Z** controls the specific system operation of a digital circuit. To improve the design the circuit designer needs to express the function as a product of maxterms. Convert the function into its maxterm representation and explain the steps involved. (CO3:8 Marks)

OR

III.B In a digital control system, a function is defined using a truth table to determine its outputs based on input combinations. To enhance the system's design, identify the Sum of Products (SOP) and Product of Sums (POS) forms of the function from the truth table.

X	Y	z	F(n)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0

1	0	1	1
1	1	0	0
1	1	1	0

(CO3:8 Marks)

BUNCH IV

IV.A.1 Imagine a computer arithmetic unit that performs binary addition as part of its processing tasks. The unit needs to add two binary digits along with a carry from a previous operation. Design a circuit using two Half Adders to implement a Full Adder, and explain how the circuit generates the correct sum and carry outputs. (CO4:8 Marks)

OR

IV.B Sophia is designing a digital home lighting system where four different rooms' lights need to be controlled using only two control switches. Each unique combination of the two switches should activate the light in one specific room while keeping the others off.

IV.B.1
i. Identify the inputs and outputs of the 2-to-4 decoder used to control the lights. (1 mark)
ii. Construct the truth table for the decoder, showing how each combination of the two control inputs corresponds to activating a specific room's light. (2.5 marks)
iii. Derive the Boolean expressions for each output based on the truth table. (2 marks)
iv. Draw the circuit diagram of the 2-to-4 decoder using basic logic gates. (2.5 marks) (CO4:8 Marks)

BUNCH V

V.A Hari is designing a digital system that requires a module to store and update a single bit of data in synchronization with a clock signal. He decides to use a **D Flip-Flop** for this purpose. Design the circuit diagram of the D Flip-Flop using basic logic gates and provide its characteristic table, showing how the output (Q) changes with the D input and the clock signal. (CO5:8 Marks)

OR

V.B There are two types of circuits- Combinational and Sequential.

V.B.1 How do you differentiate between these two types of circuits? (CO5:4 Marks)

V.B.2 Illustrate the role of a Flip Flop in storing and updating data over time. Considering JK Flip Flop as an example. (CO5:4 Marks)

CO1	Perform conversions between popular positional number systems (Decimal, Binary, Octal, Hexadecimal) and apply binary arithmetic
CO2	Analyse the working of logic gates and solve expressions using laws of Boolean algebra.
CO3	Illustrate POS and SOP expressions, simplify Boolean expressions using KMAP and implement circuits using the Universal Gates.
CO4	Illustrate the design and applications of combinational logic circuits.
CO5	Analyse sequential logic circuits, including various types of flip-flops (RS, JK, T, D), their triggering mechanisms, and the concept and functioning of registers.